

Data Science Project Preparation – Requirements

1. Project Idea

- Think of a **real-world problem or question** you are curious about.
- Your project must have a **clear objective** (e.g., "predict house prices," "analyze student study habits," "classify types of music," "detect fake news").
- The idea should be **specific and feasible**—not too broad, but still meaningful.

📎 **Deliverable:** Write a short description (3–5 sentences) explaining:

- The problem/question you want to solve.
 - Why this topic is interesting or important.
 - What type of data you think you will need.
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2. Data Collection

You are responsible for **finding or building your dataset**. Here are the methods you can use:

1. **Search online** for open datasets (Kaggle, UCI Machine Learning Repository, government open data portals, etc.).
2. **Download structured data** (CSV, JSON, Excel) that is publicly available.
3. **Collect your own data** (e.g., surveys, web scraping, personal observations).

⚠️ **Rules:**

- Dataset must have at least **200 rows (records)** and at least **5 features (columns)**.
- Data must be **relevant** to your project idea.
- Do not use datasets that are too “overused” (like the Titanic dataset) unless you add your own twist.
- If you create your own dataset, document how you collected it.

📎 **Deliverable:** Provide the raw dataset file (CSV/Excel/JSON) and the link/source where you found it (if applicable).

3. Data Documentation

For every dataset, you must explain:

1. **Source** – where you got the data.
2. **Size** – number of rows and columns.
3. **Features** – list of the columns with a short description of each.
4. **Potential Issues** – any missing values, errors, or biases you expect.

📎 **Deliverable:** A short report (1–2 pages, or a Google Doc) describing your dataset.

4. Submission Requirements

- One project idea description (short paragraph).
 - One dataset file (CSV, Excel, or JSON).
 - One short dataset documentation report.
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5. Evaluation Criteria

You will be graded on:

- **Clarity of project idea** (Is it meaningful and specific?)
 - **Quality of dataset** (Is it relevant, large enough, and useful?)
 - **Effort in data collection** (Did you go beyond just downloading the first dataset you found?)
 - **Documentation** (Did you clearly explain the dataset and its features?)
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☒ **Deadline:** [11/2/2025] ☒ **Submission method:** [Google Classroom]

6. Public Data Sources

General catalogs

- Kaggle Datasets — <https://www.kaggle.com/datasets>
- UCI Machine Learning Repository — <https://archive.ics.uci.edu/ml/index.php>
- Google Dataset Search — <https://datasetsearch.research.google.com/>
- Data.world — <https://data.world/>
- Awesome Public Datasets (GitHub index) — <https://github.com/awesomedata/awesome-public-datasets>

Government & open data portals

- Data.gov (USA) — <https://www.data.gov/>
- USAfacts — <https://usafacts.org/>
- EU Open Data Portal — <https://data.europa.eu/en>
- UK Data Service — <https://ukdataservice.ac.uk/>
- World Bank Data — <https://data.worldbank.org/>
- UN Data — <https://data.un.org/>

Education

- U.S. Dept. of Education (EDFacts) — <https://www2.ed.gov/about/inits/ed/edfacts/eden.html>
- National Center for Education Statistics (NCES) — <https://nces.ed.gov/datatools/>
- OECD Education Data — <https://stats.oecd.org/>

Health & epidemiology

- CDC Data & Stats — <https://data.cdc.gov/>
- WHO Global Health Observatory — <https://www.who.int/data/gho>
- NIH/NLM (health datasets catalog) — <https://www.nlm.nih.gov/NIHbmic/healthdata.html>

Environment, climate, air quality

- NOAA Climate Data Online — <https://www.ncdc.noaa.gov/cdo-web/>
- NASA Earthdata — <https://earthdata.nasa.gov/>
- EPA Data (Envirofacts) — <https://enviro.epa.gov/>
- OpenAQ (air quality) — <https://openaq.org/>

Economics & finance

- FRED (St. Louis Fed) — <https://fred.stlouisfed.org/>
- Bureau of Labor Statistics — <https://www.bls.gov/data/>
- Bureau of Economic Analysis — <https://www.bea.gov/data>
- SEC EDGAR (company filings) — <https://www.sec.gov/edgar/search-and-access>

Transportation & mobility

- Bureau of Transportation Statistics — <https://www.bts.gov/data>
- FAA On-Time Performance — <https://transtats.bts.gov/ONTIME/>
- NYC TLC (Taxi trips) — <https://www.nyc.gov/site/tlc/about/tlc-trip-record-data.page>

Cities & local open data (examples)

- NYC Open Data — <https://opendata.cityofnewyork.us/>
- Chicago Data Portal — <https://data.cityofchicago.org/>
- Los Angeles Open Data — <https://data.lacity.org/>

Social science & surveys

- IPUMS (census microdata) — <https://ipums.org/>
- GSS (General Social Survey) — <https://gss.norc.org/>
- Pew Research (microdata & tables) — <https://www.pewresearch.org/download-datasets/>

Sports (clean, analysis-ready)

- FiveThirtyEight Datasets — <https://github.com/fivethirtyeight/data>
- Basketball-Reference (data tables) — <https://www.basketball-reference.com/>
- Football-Data (soccer) — <https://www.football-data.co.uk/>

NLP / text / images

- Hugging Face Datasets — <https://huggingface.co/datasets>
- Project Gutenberg (texts) — <https://www.gutenberg.org/>
- Common Crawl — <https://commoncrawl.org/>
- Google Open Images (images) — <https://storage.googleapis.com/openimages/web/index.html>

Geospatial & maps

- Natural Earth — <https://www.naturalearthdata.com/>
- OpenStreetMap (planet extracts via Geofabrik) — <https://download.geofabrik.de/>
- USGS (geospatial & earth science) — <https://www.usgs.gov/products/data-and-tools>

Misc. high-quality "ready to use"

- Awesome-Public-Datasets (topic sections) — <https://github.com/awesomedata/awesome-public-datasets>
- TidyTuesday (weekly curated datasets) — <https://github.com/rfordatascience/tidytuesday>